UNIT 2: BIOSAFETY GUIDELINES

GMO(Genetically Modified Organisms):

Its genetic material has been altered using genetic engineering (by eliminating, modifying or adding copies of specific genes often from other organisms through modern biology techniques i.e, recombinant DNA- rDNA, Genetic Engineering/ Gene splicing). They are used :

Foods - Canola, Soya Bean, Golden rice, corn

Medicines - Insulin, Hormones, vaccine.

This led to concern about the dangers they might cause to the environment and human health.

Types of GM Foods

1. First-generation crops

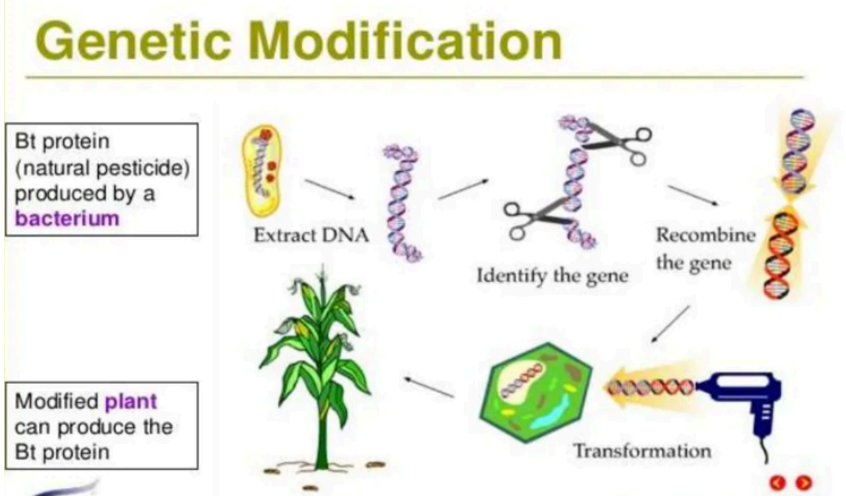
- Traits such as herbicide tolerance, better insect resistance and better tolerance to environmental stress.
- The ensuing crops are not significantly different from the traditionally grown crops in terms of appearance, taste and nutrition.
- Examples of such crops are herbicide resistant soybean, insect-resistant maize, and herbicide and insect-resistant potato.

2. Second-generation crops

- Increased levels of protein, modified or healthier fats, modified carbohydrates, increased flavor or increased micronutrients.
- Examples rice with a higher level of beta-carotene,
- tomatoes with higher levels of carotenoids,
- maize with increased vitamin C,
- soybean with improved amino acid composition, and potatoes with higher calcium content.

3. Third Generation

- Increased ability to resist abiotic stress
- Create "pharma plants" to help produce active pharmaceutical products.
- In Feb 2009, the U.S. Food and Drug Administration (FDA) approved recombinant anti-thrombin prevention of blood clots, hereditary anti-thrombin deficiency.
- This is 1st human biologic drug derived from the milk of goats that have been genetically engineered to produce human anti-thrombin in their milk.



Institutional Framework:

rDNA Advisory committee

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Institutional Biosafety Committees (IBSC) Department of Biotechnology, Ministry of Science and Technology.

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Review Committee on Genetic Manipulation (RCGM) Department of Biotechnology

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Genetic Engineering Approval Committee (GEAC) under the Ministry of Environment and Forests

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State Biotechnology Coordination Committee (SBCC)

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District Level Committee (DLC) in the districts

Trait	Advantage	Sample Product
Pest-Resistance	Less damage by insect, virus, bacteria, etc.	Corn
Herbicide-Resistance	Hebicides will kill only weeds, not crops	Cotton
Delayed Ripening	Can be shipped with less damage	Tomato
Miniature Size	Improved eating quality	Watermelon
Improved Sweetness	Better tasting	Sweet peas
Cold-Resistance	Withstands freezing and thawing	Strawberries
High Starch	Absorbs less oil when fried	Potato
Polyester Gene Added	Better fiber properties	Cotton
Growth Hormone Added	Faster growth	Salmon
Hepatitis B Virus Protein Added	May provide immunity to Hepatitis	Banana

The Role of IBSC

Institutional Biosafety Committee (IBSC) shall be constituted by all institutions handling hazardous microorganisms and/or GE organisms. The Committee will be the nodal point for implementation of the biosafety guidelines and for the interactions within the institution.This is a small committee established by every institution engaged in rDNA research and the related productionactivities.It monitors rDNA research activities at the institutional level.

The Committee shall comprise the Head of the Institution, Scientists engaged in the recombinant DNA work, a medical doctor and a nominee of the Department of Biotechnology.

- Sends report to RCGM regarding observance of safety guidelines on accidents risks and on deviations if any
- Reviews the requirements of guidelines for safety of new projects
- Allows some person to take training on biosafety in research activities
- Takes emergency plans in urgent situations
- Attempts to provide medical care to people working in the laboratory

The constitution, composition, role and functions, information for compliance requirements, processes to be followed while dealing with hazardous microorganisms, GE organisms or cells and product thereof in line with Rules 1989 is described in “Guidelines and Handbook for Institutional Biosafety Committee”. Adherence to the guideline shall be binding for all IBSCs.

Genetic Engineering Appraisal Committee (GEAC)

Genetic Engineering Appraisal Committee (GEAC) [formerly known as Genetic Engineering Approval Committee (GEAC) name changed through Gazette notification, G.S.R No. 613 dated 16th July 2010], has been established under the Ministry of Environment, Forest and Climate Change (MoEF&CC). The committee or any persons authorized by it has powers to take punitive action under the Environment (Protection) Act, 1986. GEAC gives approval for :

- Import, export, transport,production and sale of GEOs and other organisms
- Release of GEOs from the laboratories to environment
- Large scale culture and use of GEOs and microbes in industries
- Use of GEOs in field application and experimental trials
- Monitoring the risks and accidents due to GEOs

Review Committee on Genetic Manipulation (RCGM)

This Committee shall function from the Department of Biotechnology to monitor the safety related aspect in respect of on-going research projects or activities involving hazardous microorganisms, GE organisms and cells and products thereof. The RCGM shall include representatives of (a) Department of Biotechnology (b) Indian Council of Medical Research (c) Indian Council of Agricultural Research (d) Council of Scientific and Industrial Research (e)

other experts in their individual capacity. RCGM may appoint sub groups to assist RCGM on matters related to risk(s) assessment, in reviewing existing and preparing new guidelines.

Functions:

- Establishes the procedural guidance manual for regulatory process with GEOs
- It reviews the risk potentials of GEOs in the laboratory and field experiments
- It decides which containment have to be followed for experiments with risky hazardous microbes
- It advices custom authorities on import of GEOs and other biological materials from other nations
- It provides advice on IPR and patents

TERMS OF REFERENCE (TOR) OF RCGM

As per the Rules for the manufacture, use, import, export & storage of hazardous microorganisms, genetically engineered organisms or cells, 1989 notified under the Environment (Protection)Act, 1986 and as amended from time to time, the Review Committee on Genetic Manipulation (RCGM) shall function in the Department of Biotechnology (DBT) .

1. RCGM shall monitor the safety related aspects in respect of ongoing recombinant DNA (r-DNA) projects and activities involving Genetically Engineered (GE) organisms/ hazardous microorganisms. All ongoing projects including high risk category and confined field experiments shall be reviewed by RCGM to ensure that adequate precautions and containment conditions are complied with as per the Guidelines and Standard Operating Procedures (SOPs) issued by DBT from time to time.
2. The RCGM shall lay down procedures restricting or prohibiting production, sale, importation and use of such GE organisms or products thereof for research and applications as mentioned in the schedule of Rules, 1989.
3. RCGM shall bring out manuals and guidelines specifying procedure for regulatory processes with respect to activities involving GE organisms in research, use and application including industry with a view to ensure safety to human, animal, and environment health.
4. The RCGM would issue the clearance letters/permits for import or exchange of genes, DNA fragments, vectors, plasmids, cosmids, etiologic agents and transgenic organisms or germplasm(s) including transformed calli, seeds, plants and plant parts for research use only. It will also take note of all such commercially available agents which are acquired from commercial sources through Institutional Biosafety Committee (IBSC).
5. The RCGM shall be the regulatory body for receiving and reviewing the applications to conduct confined field trials (such as event selection trials, Biosafety Research Level I trials (BRL-I), pollen flow studies or any other trial involving GE organisms) and recommend appropriate studies to be conducted for data generation for biosafety assessment as per clause 4, as per the decision of the Genetic Engineering Approval Committee (GEAC) for its authorization.

6. The RCGM, on case-by-case basis, can authorize applicants to use bioreactors to produce sufficient material/end-products of GE organisms required for conducting pre-clinical studies and other relevant data generation including the use of capacity over and above as specified by Recombinant DNA Safety Guidelines, 1990. It will authorize to produce sufficient quantities of GE organism/plant product(s) for generating safety data in appropriate animals as per National and International Guidelines, as appropriate on case-by-case basis.
7. The RCGM, if required, call generate, examine or invite the research projects, proposals for capacity building and training courses in biosafety, creation of information systems/data banks in electronic media, websites etc. for financial support and recommend the same to DBT for furthering the cause of generating specific biosafety data related to use of GMOs and strengthening infrastructure facilities & dissemination of information on biosafety rules, regulations and guidelines in the country.
8. The RCGM may also invite, induct or appoint special experts in their individual capacities on a case-by-case basis.
9. The RCGM may constitute subgroups of its members or subcommittees involving experts for specific functions or seek advice on specific matters from other external experts.
10. The RCGM or its constituted subgroups/subcommittees may visit periodically the experimental sites where r-DNA projects and activities involving GE organisms/ hazardous microorganisms are being pursued to ensure that adequate safety measures have been taken as per the guidelines and compliance of SOPs.
11. The RCGM shall maintain the classified/confidential business information provided by the applicants as confidential.
12. The RCGM shall function for a period of three years from the date of notification.
13. The RCGM shall meet once in a month or as and when required.
14. The members of RCGM shall be paid TA/DA and honorarium as per the Government of India rules as applicable from time to time.

Cartagena Protocol

The Cartagena Protocol on Biosafety to the Convention on Biological Diversity is an international treaty governing the movements of Living Modified Organisms (LMOs) resulting from modern biotechnology from one country to another.

- seeks to protect biodiversity from the potential risks caused by LMOs arising from modern technology.
- The protocol was adopted because of the tremendous advancements in biotechnology and the associated concerns about its safety and usage with respect to biodiversity.
- It seeks to implement an internationally harmonized regime for biosafety in order to ensure the safe utilization of modern biotechnology.
- The Protocol has provisions for an **Advance Informed Agreement (AIA)** procedure ensuring that countries are given enough information to make informed decisions before agreeing to import LMOs into their country.

- There are four components to the AIA:
    - Notification by the exporter (This is a detailed written description of the LMO by the exporter, well in advance of the first shipment)
    - Acknowledgement of notification receipt by the importer
    - Decision procedure (Approve/prohibit/ask for more information, etc.)
    - Review of decisions
- sets up a Biosafety Clearing-House (BCH) to enable information exchange on LMOs between countries. The BCH is an information-sharing mechanism for relevant technical, scientific and legal information.
- gives a precautionary approach to the issue of transfer of LMOs from one country to another.
- The Cartagena Protocol chiefly governs the following:
    - LMOs that are intentionally introduced into the environment (trees, seeds or fish).
    - GM farm commodities (grain and corn used for animal feed, food or for processing).
    - does not cover pharmaceuticals for humans addressed by other international agreements and organizations or products derived from LMOs, such as cooking oil from GM corn.
- LMOs are classified as the following under the Protocol:
    - LMOs for intentional introduction into the environment – subject to AIA procedures.
    - LMOs for direct usage as food or feed, or for processing – subject to simplified procedures which include informing through the BCH.
    - LMOs for contained usage (like bacteria for lab experiments) – these are exempt from AIA procedures.
- **India** is a party to the Cartagena Protocol (ratified in 2003). The nodal agency (Competent National Authority-CNA) in the country for the implementation of the Protocol is the Ministry of Environment, Forest and Climate Change (MOEF&CC), Government of India.
    - Regarding setting up of procedures for regulating LMOs, India was one of the early movers in the development of a biosafety regulatory framework, way back in 1989, and has a systematic and structured science-based regulatory system.
    - In the Indian regulations, the terms Genetically Engineered Organism or Genetically Modified Organism are used, which are synonymous with LMOs.
    - In India, series of guidelines are available for risk assessment and risk management of GMOs.

Quiz:

1. What is biosafety committee and its functions?

   A biosafety committee, also known as an Institutional Biosafety Committee (IBC), reviews and oversees activities that involve biological agents and biohazards.

   Functions:

- **To review periodically the safety and control measures established at various institutions handling GE organisms.**
- **To act as a nodal agency at the State level to assess the damage, if any, due to the release of GE organisms and to take on-site control measures.**

2. Expand the term LMO. **Living Modified Organisms**

3. The physical containment include: **Air filtration, sterilization lights, waste disposal, protective handling**